



Antigravity Directive: Capital Allocation, Multi-Pool Seeding & Omnichannel Bot

Target Network: Arbitrum Sepolia (421614)

Context: The bot wallet currently holds exactly **3.766 ETH, 15,969.708 USDC, and 0.0378 WETH**. We need to strategically allocate this capital to kickstart the beta test, seed the three physical Uniswap V3 pools, keep a reserve, and load the 3-3-1 Waterfall with early yield.



Task 1: The Asset Allocation Strategy

Before seeding the pools, the script `scripts/seed-testnet-pools.ts` must route the existing bot wallet funds exactly as follows:

1. **The Waterfall Seed (10,000 USDC):** * Approve and call `notifyRewardAmount(10000 * 10**USDC_DECIMALS)` on the `GuildDistributor` contract. This immediately loads the triple-accumulator with a massive reward pool so early beta testers see skyrocketing APYs as soon as they stake.
2. **The Protocol Reserve (1,000 USDC):**
 - Keep 1,000 USDC in the bot wallet untouched for future gas paymaster refills or emergency liquidity.
3. **The Pool Capital (4,969 USDC + ETH/WETH):**
 - The remaining ~4,969 USDC, along with a portion of the 3.766 ETH (which will need to be wrapped into WETH), will be utilized in Task 2 to seed the three DEX pools.



Task 2: The Multi-Pool Seeding Script

In `scripts/seed-testnet-pools.ts`, execute the following actions using the Uniswap V3 Router and NonfungiblePositionManager (NFPM: `0xC36442b4a4522E871399CD717aBDD847Ab11FE88`).

Step 1: Acquire USDT for the Stable Pool

- The bot wallet currently has no USDT. Use the Uniswap V3 Router to swap exactly 1,000 USDC into USDT.

Step 2: Mint the 3 Liquidity Positions (Wide/Full Tick Range)

1. **The Safe Vault (USDC/USDT - 0.01% or 0.05% Fee Tier):** * Seed with the newly swapped 1,000 USDT and 1,000 USDC.
2. **The Medium Vault (WETH/USDC - 0.3% Fee Tier):** * Seed with the remaining 2,969 USDC and the equivalent WETH value.
 - **CRITICAL:** The wallet only holds 0.0378 WETH. The script must execute a `deposit()` on the WETH contract to wrap enough of the raw ETH to cover this pairing

(approximately ~0.85 ETH depending on the current testnet oracle price).

3. **The Degen Vault (GUILD/WETH - 1% Fee Tier):** * Wrap an additional 1.0 ETH into WETH.
 - Seed this pool with 50,000 GUILD and 1.0 WETH (assuming a 1:1 relative initialization price).

Note: Ensure YieldVault takes ownership of the resulting ERC-721 position NFTs so it can securely manage the harvesting of fees.



Task 3: Upgrading to the Omnichannel Volume Bot

Update volume-bot.ts into a multi-route loop that generates fees across the entire ecosystem using the remaining wallet reserves (~1.8+ raw ETH).

1. The Multi-Route Ping-Pong Loop

Refactor the bot to loop through this exact sequence every few minutes:

- **Route 1 (Stables):** Swap 10 USDC to USDT, then swap exact USDT back to USDC.
- **Route 2 (Blue-Chips):** Swap 0.01 WETH for USDC, then swap exact USDC back to WETH.
- **Route 3 (Degen):** Swap 0.01 WETH for GUILD, then swap exact GUILD back to WETH.

The Result: By swapping back and forth across all three pairs, the bot pays the pool fees on both sides. This rapidly fills all three physical testnet pools with fee revenue without exposing the bot to heavy directional market risk.

2. The Harvest Trigger

Every 10th loop, have the bot call YieldVault.harvestAndDistribute() to pull the accrued fees from all active positions, swap necessary pairs to USDC via the router, and push them into the GuildDistributor Waterfall.



Verification Steps

1. Run `npx tsx scripts/seed-testnet-pools.ts`. Verify the NFPM mints all three ERC-721 liquidity positions successfully on Arbiscan and the Distributor receives the 10k USDC payload.
2. Run `npx tsx volume-bot.ts`. Monitor the terminal to ensure it executes swaps across all three pool routes.
3. Open the frontend to the Vault page. The 10k USDC seed should immediately reflect in the APYs, and the bot's volume should continuously tick the "Total Pending Yield" upward.